

$m_1 = 250$

Supervised Learning

- Model trained using labeled Data
- correct in/out classification

Regression

Ex: "House price"

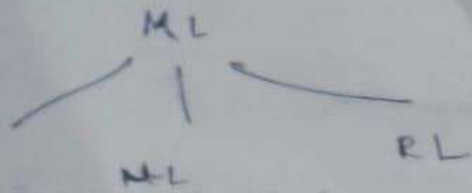
unsupervised Learning

- model trained using unlabeled Data

- No correct output

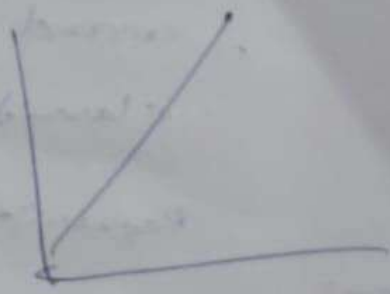
03-08-26

AI - Agent



Random process

- SVM
- KNN
- XG Boost



$$y = mx + c$$

x_1	x_2	y
1000	5	500000
2000	10	1000000

$$M_1 \times 1000 + M_2 \times 5 = 500000$$

$$M_1 \times 2000 + M_2 \times 10 = 1000000$$

LLM

- LSTM - long short term memory
- RNN (Recurrent Neural Network)
 - Short - term memory
 - Reads every word in order of sequence
 - Remembers previous information while reading process the sequence

Attention Mechanism:

word 1 love AI.

[1 , love , AI ,] - tokens

↓

[1 , 2 , 3] ~~tokens~~ Tokens

ob₁

ob₂

ob₃

Embedded
matrix

Numerical Representation

157M

- Improved Version of RNN
- Remembers important information for much longer

Gates:

- Forget Gate - Decide what old information to discard
- Input Gate - Decide what new info. to be stored
- Output Gate - Decide what information to pass to next step

- Attention Mechanism allows the model to directly look back at important words instead of relying only on previously
- Attention Score based

Calculate Score

convert to probability

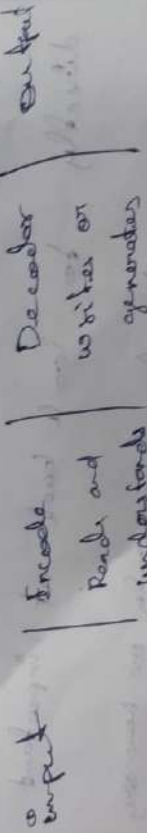
Focus on Important word

Softmax function converts the

Score into probabilities that add up to

$(Q, k, v) - (Query, Key, Value)$

Transformer



Common Algorithm

classification

• Random Forest

• SVM

• KNN

Regression

• Linear Regression

• Decision Tree

Cross validation

Drop out

Early stopping

Regularization

Data Augmentation

Normalization

Hyper parameter tuning

Embedding

raw word $\rightarrow$ input token

converted into a vector

best numerical representation

Positional Encoding

positions the words order

1 - pos 1

love - pos 2

is - pos 3

Architecture:

Input

Embedding

Positional Encoding

Encoder

Decoder

Linear Attention

softmax

output